Dylan Frith

6/18/26

1. Automotive Tire Sales Modeling

Using the data from `TireSales.csv`, answer the following questions.

```
Tires_df = read.csv("https://www.businessregression.com/Data/TireSales.csv")
```

a. Fit a linear model to predict the response variable (Sales) from the variables: `CompPrice`, `Income`, `Ads`, `Cars`, `Price` and `Age`. Display a summary of the model.

```
reg = lm(Sales ~ CompPrice + Income + Ads + Cars + Price + Age, data = Tires_df)
summary(reg)
```

```
##
## Call:
## lm(formula = Sales ~ CompPrice + Income + Ads + Cars + Price +
##     Age, data = Tires_df)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.0340 -1.2340  0.0302  1.3632  5.3442
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.6298913  1.2007986   2.190   0.0290 *
## CompPrice    0.0992546  0.0188199   5.274 2.00e-07 ***
## Income      -0.0003679  0.0029141  -0.126   0.8996
## Ads          0.0376981  0.0149658   2.519   0.0121 *
## Cars         8.0822521  0.1921791  42.056  < 2e-16 ***
## Price       -0.0966082  0.0153164  -6.308 6.31e-10 ***
## Age         -0.0157475  0.0129140  -1.219   0.2233
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.786 on 493 degrees of freedom
## Multiple R-squared:  0.7949, Adjusted R-squared:  0.7924
## F-statistic: 318.4 on 6 and 493 DF,  p-value: < 2.2e-16
```

b. Clearly state the regression equation.

$$\hat{y} = 2.62989 + 0.09925x_1 - 0.000368x_2 + 0.037698x_3 + 8.08225x_4 - 0.096608x_5 - 0.01575x_6$$

c. Calculate the VIF values corresponding to each predictor variable.

```
library(car)
```

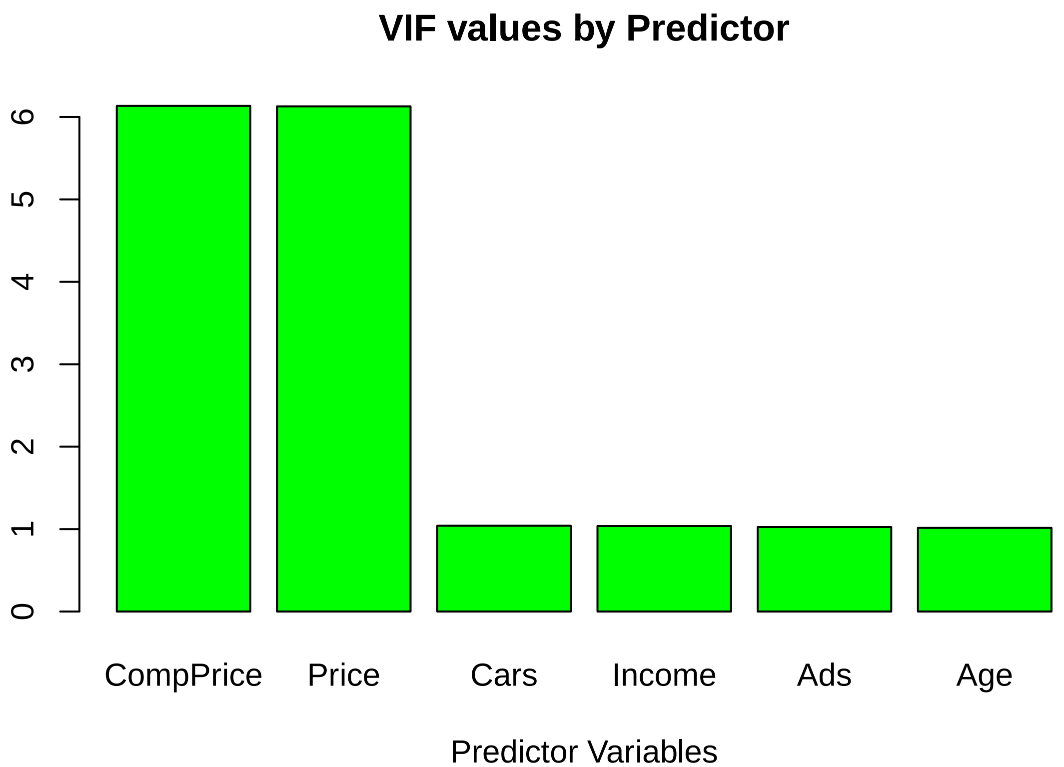
```
## Loading required package: carData
```

```r
vif_values = vif(reg)
vif_values
```

```
## CompPrice    Income       Ads      Cars     Price       Age
##  6.134710  1.037864  1.025674  1.040806  6.128783  1.014830
```

d. Construct a bar graph using the VIF values calculated above.

```r
sorted_vif_values <- vif_values[order(-vif_values)]
barplot(sorted_vif_values, col= "green", xlab = "Predictor Variables", main = "VIF values by Predictor")
```

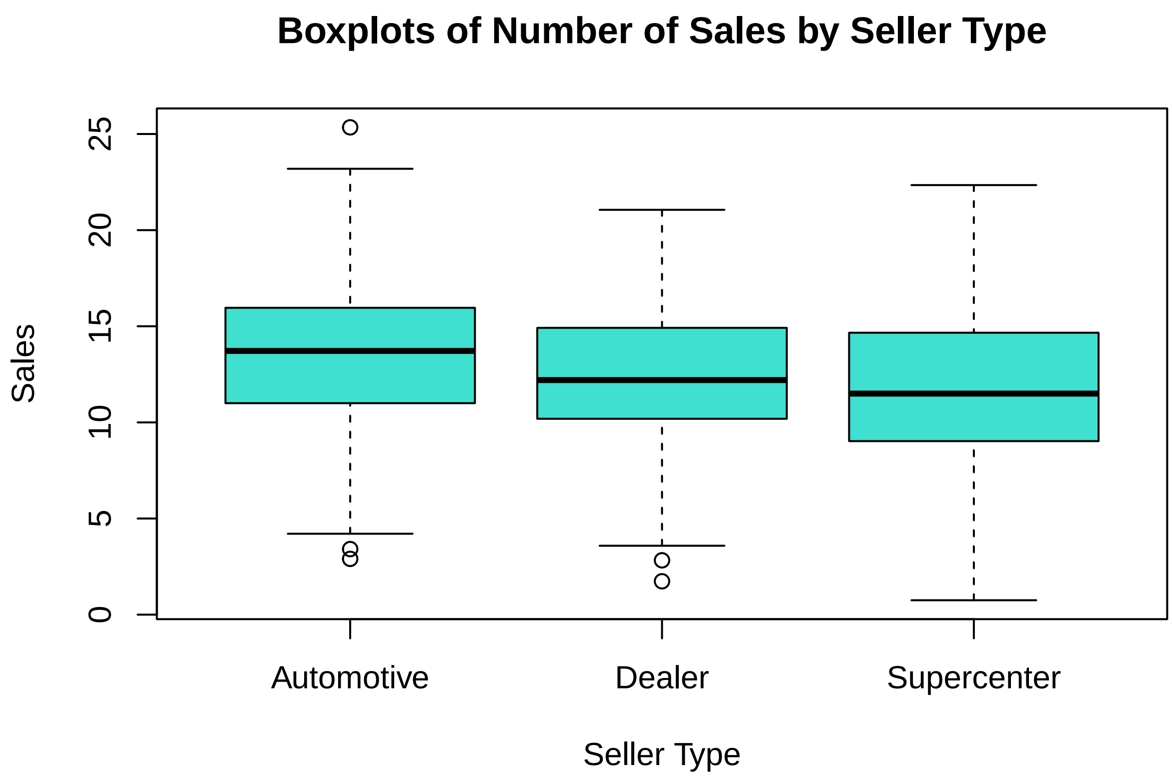


2. Automotive Tire Sales by Seller Type

Using the data from `TireSales.csv`, answer the following questions.

a. Construct boxplots of the number of sales by seller type.

```r
boxplot(Sales ~ SellerType, data = Tires_df, col = "turquoise", xlab = "Seller Type", ylab = "Sales", ma
```



b. It is believed that sales are significantly different across the different seller types. Clearly state the null and alternative hypotheses.

$$H_O : \mu_1 = \mu_2 = \mu_3$$
$$H_A : \mu_1 \neq \mu_2 \neq \mu_3$$

c. Calculate the F statistic.

```
reg2 = lm(Sales ~ SellerType, data = Tires_df)
anova(reg2)
```

```
## Analysis of Variance Table
##
## Response: Sales
##             Df Sum Sq Mean Sq F value    Pr(>F)
## SellerType   2  296.9 148.455  10.008 5.483e-05 ***
## Residuals  497 7372.7  14.834
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d. Can you refute the null hypothesis using an F test with a significance level of $\alpha = 0.05$?

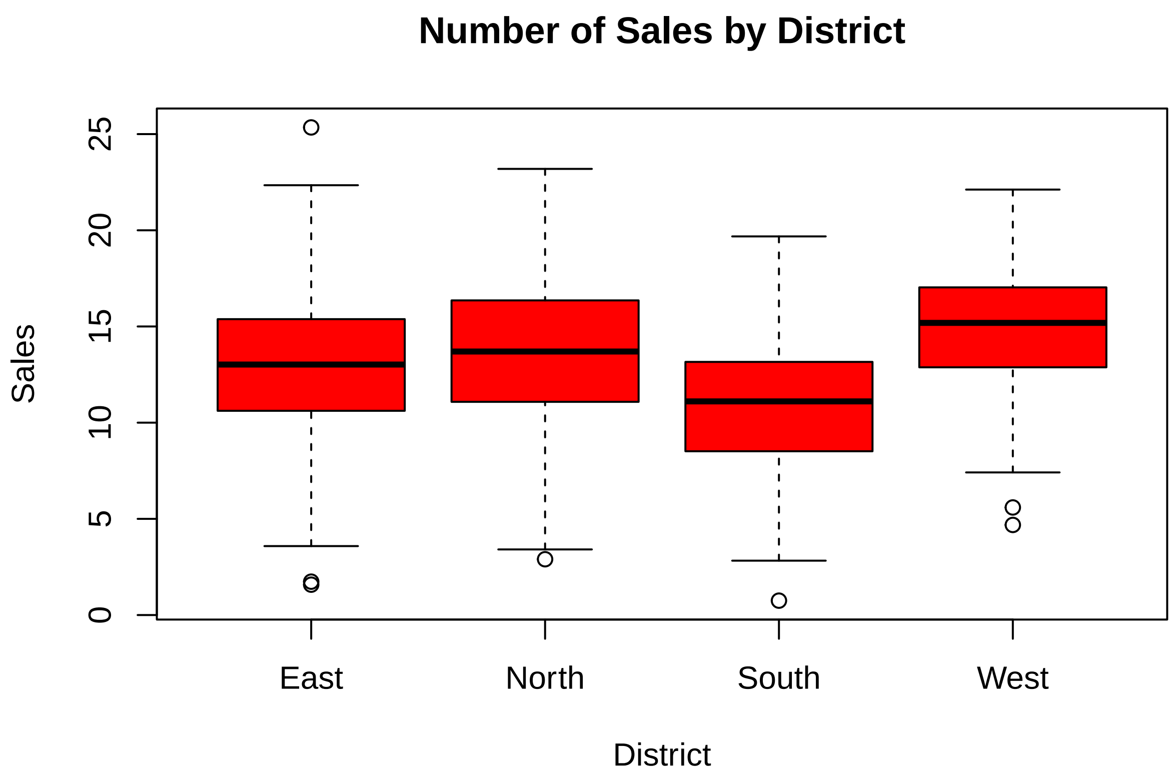
At significance level of alpha = 0.05, we reject the null since the p-value < 0.05.

3. Automotive Tire Sales by District

Using the data from `TireSales.csv`, answer the following questions.

a. Construct boxplots of the number of sales by district.

```
boxplot(Sales ~ District, data = Tires_df, col = "red", xlab = "District", ylab = "Sales", main = "Numb
```



b. It is believed that sales are significantly different across the different districts. Clearly state the null and alternative hypotheses.

$$H_O : \mu_1 = \mu_2 = \mu_3 = \mu_4$$
$$H_A : \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$$

c. Calculate the F statistic.

```
reg3 = lm(Sales ~ District, data = Tires_df)
anova(reg3)
```

```
## Analysis of Variance Table
##
## Response: Sales
##            Df Sum Sq Mean Sq F value   Pr(>F)
## District    3  567.8 189.258  13.218 2.58e-08 ***
## Residuals 496 7101.8  14.318
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

d. Can you refute the null hypothesis from the F test with a significance level of $\alpha = 0.05$?

At significance level of alpha = 0.05, we reject the null since the p-value < 0.05

4. Coral Gables Housing

Jigar is interested in purchasing a house in Coral Gables and wants to get the largest total living area for the price. He collects the size and sale price of 5 houses in a particular Coral Gables neighborhood. The data shown here reflect the size (in square feet) and the sale price (in thousands of dollars).

```r
Size = c(2700, 3500, 2600, 5300, 4200)
Price = c(650, 875, 670, 1200, 981)
```

Use the Coral Gables housing data from the previous problems and do the following using R.

a. Print a summary of the regression equation that predicts price as a function of house size.

```r
reg4 = lm(Price ~ Size)
summary(reg4)
```

```
##
## Call:
## lm(formula = Price ~ Size)
##
## Residuals:
##       1       2       3       4       5
## -30.703  32.216   9.557  -7.466  -3.605
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 133.68052   45.10758   2.964  0.05937 .
## Size          0.20260    0.01188  17.048  0.00044 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 26.71 on 3 degrees of freedom
## Multiple R-squared:  0.9898, Adjusted R-squared:  0.9864
## F-statistic: 290.6 on 1 and 3 DF,  p-value: 0.0004397
```

b. Use the regression equation to predict the price when the size of the house is 2500 square feet. **Hint:** `obs_x` has been given.

```r
obs_x = data.frame(Size = 2500)
predict(reg4, data.frame(Size = 2500))
```

```
##        1
## 640.1829
```

c. Find a 95% confidence interval for the price when the size of the house is 2500 square feet.

```r
predict(reg4, data.frame(Size = 2500), interval = "confidence")
```

```
##        fit      lwr      upr
## 1 640.1829 582.1296 698.2362
```

d. Find a 95% prediction interval for the price when the size of the house is 2500 square feet.

```r
predict(reg4, data.frame(Size = 2500), interval = "prediction")
```

```
##        fit      lwr      upr
## 1 640.1829 537.2417 743.1241
```

5. Coral Gables Housing ANOVA

Using the Coral Gables Housing data, do the following calculations without generating the ANOVA table (you will generate the ANOVA table in the next question).

a. Calculate the SSR.

```r
yhat = reg4$fitted.values
SSR = sum((yhat - mean(Price))^2)
SSR
```

```
## [1] 207370.2
```

b. Calculate the SST.

```r
SST = sum((Price - mean(Price))^2)
SST
```

```
## [1] 209510.8
```

c. Calculate the MSR.

```r
MSR = SSR/1
MSR
```

```
## [1] 207370.2
```

d. Calculate the F statistic.

```r
n = length(Size)
SSE = SST - SSR
MSE = SSE/(n-2)
F = MSR/MSE
F
```

```
## [1] 290.6212
```

6. Coral Gables Housing ANOVA in R

Using the Coral Gables Housing data, do the following using R.

a. Find the p-value of the F statistic calculated in the previous problem using the `pf` function.

```r
1-pf(F,1,n-2)
```

```
## [1] 0.0004396693
```

b. Use the `anova` function to construct the ANOVA table.

```r
anova(reg4)
```

```
## Analysis of Variance Table
##
## Response: Price
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Size       1 207370  207370  290.62 0.0004397 ***
## Residuals  3   2141     714
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```